

Networking Protocols

Chapter 1

What is a Network Protocol?

A Network Protocol is a set of rules and standards that define how data is transmitted and received between devices on a network. Protocols ensure reliable, secure, and error-free communication between computers and other network devices.

Why are Protocols Important?

Protocols are important because they:

- Enable Communication Between Devices
- Ensure Accurate Data Transfer
- Provide Error Detection
- Improve Network Security
- Standardize Communication
- Support Different Network Services

Types of Network Protocols

The major networking protocols are:

- | | |
|---------------------------------------|--|
| • TCP (Transmission Control Protocol) | • HTTPS (Hypertext Transfer Protocol Secure) |
| • IP (Internet Protocol) | • FTP (File Transfer Protocol) |
| • UDP (User Datagram Protocol) | • SMTP (Simple Mail Transfer Protocol) |
| • HTTP (Hypertext Transfer Protocol) | • DNS (Domain Name System) |

How Network Protocols Work

When data is sent over a network:

- ① The sender divides the data into packets.
- ② The packets are transmitted using network protocols.
- ③ The receiver collects and reassembles the packets.
- ④ Error checking ensures the data is received correctly.

Advantages of Network Protocols

- Reliable Communication
- Faster Data Transfer
- Secure Information Exchange
- Interoperability Between Devices
- Efficient Network Management

Applications of Network Protocols

- Internet Browsing
- Email Communication
- File Sharing
- Video Streaming
- Online Gaming
- Cloud Computing

Summary

Network protocols are the foundation of computer networking. They define the rules for communication between devices and ensure secure, reliable, and efficient data transmission across local and global networks.

Networking Protocols

Chapter 2 : TCP (Transmission Control Protocol)

What is TCP?

TCP (Transmission Control Protocol) is a connection-oriented protocol that provides reliable, ordered, and error-free delivery of data between two devices over a network.

Key Features of TCP

- Connection-Oriented Protocol
- Reliable Data Transfer
- Ensures Data Integrity
- Supports Flow Control
- Supports Congestion Control
- Full-Duplex Communication

How TCP Works?

- ① A connection is established between sender and receiver using the three-way handshake.
- ② Data is divided into segments and sent.
- ③ Each segment has a sequence number.
- ④ The receiver sends an acknowledgement (ACK) for received segments.
- ⑤ If a segment is lost or corrupted, TCP retransmits it.
- ⑥ After data transfer, the connection is closed using four-way handshake.

Three-Way Handshake (Connection Establishment)

- ① Sender sends SYN packet to receiver.
- ② Receiver sends SYN-ACK packet in response.
- ③ Sender sends ACK packet to establish the connection.

Four-Way Handshake (Connection Termination)

- ① One side sends FIN packet.
- ② Other side sends ACK.
- ③ Other side sends FIN.
- ④ First side sends ACK and connection is closed.

TCP Header Format

TCP header contains source port, destination port, sequence number, acknowledgement number, flags, window size, checksum, etc.

Advantages of TCP

- Provides reliable and error-free data transfer.
- Maintains the order of data.
- Supports flow and congestion control.
- Suitable for applications that require accuracy.

Disadvantages of TCP

- More overhead due to connection setup and acknowledgements.
- Slower than UDP because of reliability mechanisms.
- Not suitable for real-time applications.

Applications of TCP

TCP is used in web browsing (HTTP/HTTPS), email (SMTP), file transfer (FTP), remote login (Telnet/SSH) and other applications that require reliable communication.

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Chapter 3 : IP (Internet Protocol)

What is IP?

IP (Internet Protocol) is a connectionless protocol responsible for routing and addressing packets so that they can travel across networks and reach the correct destination.

Key Features of IP

- Connectionless protocol
- Logical addressing using IP addresses
- Best-effort delivery (no guarantee of delivery)
- Supports routing across multiple networks
- Works at the Network Layer (Layer 3 of OSI model)

IP Address

An IP address is a unique numerical label assigned to each device on a network.

- IPv4 Address : 32-bit address (e.g., 192.168.1.1)
- IPv6 Address : 128-bit address (e.g., 2001:0db8:85a3::8a2e:0370:7334)

Types of IP Addresses (IPv4)

- Class A : 1.0.0.0 to 126.255.255.255
- Class B : 128.0.0.0 to 191.255.255.255
- Class C : 192.0.0.0 to 223.255.255.255
- Class D : 224.0.0.0 to 239.255.255.255 (Multicast)
- Class E : 240.0.0.0 to 255.255.255.255 (Reserved)

IP Packet Format (IPv4)

An IP packet contains the following fields:

- | | | |
|-------------------|----------------------|--------------------------|
| • Version | • Flags | • Source IP Address |
| • Header Length | • Fragment Offset | • Destination IP Address |
| • Type of Service | • Time to Live (TTL) | • Options (if any) |
| • Total Length | • Protocol | • Data (Payload) |
| • Identification | • Header Checksum | |

How IP Works?

- ① The sender creates an IP packet with source and destination IP addresses.
- ② Routers examine the destination IP and decide the best path.
- ③ The packet is forwarded from one router to another.
- ④ The packet reaches the destination network.
- ⑤ Finally, the packet is delivered to the destination device.

Advantages of IP

- Enables communication across different networks.
- Supports routing and data forwarding.
- Scalable and widely used in the Internet.
- Supports both IPv4 and IPv6.

Applications of IP

IP is used in almost all internet services including web browsing, email, file transfer, video streaming, online gaming, and more.

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Chapter 4 : UDP (User Datagram Protocol)

What is UDP?

UDP (User Datagram Protocol) is a connectionless transport layer protocol that allows fast data transmission without establishing a connection.

It is simple, lightweight, and suitable for real-time applications.

Key Features of UDP

- Connectionless Protocol
- No Handshake (No connection establishment)
- Faster than TCP
- No Guarantee of Delivery
- Does not ensure order of packets
- Low overhead
- Suitable for real-time applications

How UDP Works?

- ① The sender sends data as UDP datagrams to the receiver.
- ② No connection is established between sender and receiver.
- ③ UDP adds a header to the data (no complex control).
- ④ Receiver receives the datagram and passes it to the application.
- ⑤ No acknowledgement is sent back to the sender.

UDP Header Format

UDP header is only 8 bytes long and contains 4 fields.

Field	Size (bits)	Description
Source Port	16	Port number of the sending application
Destination Port	16	Port number of the receiving application
Length	16	Length of UDP header + data
Checksum	16	Error checking to detect corruption

Advantages of UDP

- Faster than TCP due to no connection setup
- Low latency and low overhead
- Suitable for real-time applications (gaming, streaming, VoIP)
- Simple and efficient
- Supports broadcast and multicast

Disadvantages of UDP

- No guarantee of delivery
- No flow control
- No error recovery
- Packets may arrive out of order
- Less reliable than TCP

Applications of UDP

- Online Gaming
- Video Streaming
- VoIP (Voice over IP)
- Live Broadcasting
- DNS (Domain Name System)
- DHCP (Dynamic Host Configuration Protocol)
- TFTP (Trivial File Transfer Protocol)
- Network Monitoring and Management

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Chapter 5 : HTTP and HTTPS

What is HTTP?

HTTP (HyperText Transfer Protocol) is an application layer protocol used for transferring data on the World Wide Web. It defines how messages are formatted and transmitted and how web servers and browsers should respond to various commands.

Key Features of HTTP

- Client-Server Protocol
- Stateless (each request is independent)
- Uses TCP (usually on port 80)
- Supports different request methods (GET, POST, PUT, DELETE, etc.)
- Used for web browsing and data transfer

Working of HTTP

- ① The client (browser) sends an HTTP request to the server.
- ② The server processes the request.
- ③ The server sends an HTTP response back to the client.
- ④ The client displays the content.

Common HTTP Methods

Method	Description
GET	Requests data from the server.
POST	Sends data to the server to create/update a resource.
PUT	Updates an existing resource on the server.
DELETE	Deletes a resource from the server.
HEAD	Requests the header information without the body.

What is HTTPS?

HTTPS (HyperText Transfer Protocol Secure) is the secure version of HTTP. It uses SSL/TLS encryption to secure data between the client and server.

Key Features of HTTPS

- Uses encryption for secure communication
- Prevents eavesdropping and data tampering
- Uses TCP port 443
- Provides authentication using SSL/TLS certificates

Differences Between HTTP and HTTPS

Feature	HTTP	HTTPS
Full Form	HyperText Transfer Protocol	HyperText Transfer Protocol Secure
Security	Not Secure (No encryption)	Secure (Uses SSL/TLS encryption)
Port	80	443
URL	http://	https://
Data Protection	No	Yes
Performance	Faster	Slightly slower due to encryption

Applications of HTTP and HTTPS

- Web Browsing
- Web APIs and Web Services
- E-commerce Websites
- Online Banking and Secure Transactions
- Email Services and Cloud Applications

Summary

HTTP is used for standard web communication, while HTTPS provides secure and encrypted communication to protect sensitive data over the internet.

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Chapter 6 : FTP and SFTP

What is FTP?

FTP (File Transfer Protocol) is a standard network protocol used to transfer files between a client and a server over a TCP based connection. It allows users to upload, download, delete, rename and manage files on a remote server.

Key Features of FTP

- Uses TCP (Port 21 for control connection and Port 20 for data connection)
- Supports authentication using username and password
- Allows file upload and download
- Supports directory listing and file management
- Works in client-server model
- Plain text protocol (not secure)

How FTP Works?

- ① The client connects to the server using TCP port 21.
- ② The client sends login credentials (username and password).
- ③ After successful login, the client can issue commands.
- ④ For file transfer, a separate data connection is established (port 20).
- ⑤ Files are transferred between client and server.

Common FTP Commands

Command	Description
USER	Specifies the user name.
PASS	Specifies the user password.
LIST	Lists files and directories in the current directory.
RETR	Downloads a file from the server.
STOR	Uploads a file to the server.
DELE	Deletes a file on the server.
QUIT	Closes the connection.

What is SFTP?

SFTP (SSH File Transfer Protocol) is a secure file transfer protocol that uses SSH (Secure Shell) to provide encrypted communication between client and server.

Key Features of SFTP

- Uses SSH for secure file transfer
- All data (including login credentials) is encrypted
- Uses TCP port 22
- Provides authentication and data integrity
- Supports file upload, download and file management

FTP vs SFTP

Feature	FTP	SFTP
Protocol	File Transfer Protocol	SSH File Transfer Protocol
Security	Not Secure (Plain Text)	Secure (Encrypted)
Port	21 (Control), 20 (Data)	22
Authentication	Username & Password (Plain Text)	Username & Password (Encrypted)
Data Transfer	Unencrypted	Encrypted
Best Used For	Non-sensitive data transfer	Secure and sensitive data transfer

Applications of FTP and SFTP

- Website file management and uploads
- Software distribution
- Backup and data transfer between systems
- Secure file sharing over a network

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Chapter 7 : SMTP, POP3 and IMAP

What is SMTP?

SMTP (Simple Mail Transfer Protocol) is an application layer protocol used for sending email messages from the sender's mail client to the recipient's mail server.

Key Features of SMTP

- Used for sending emails.
- Works on TCP port 25 (or 587 for submission) and 465 for SMTPS.
- Supports authentication (using AUTH command).
- Transfers email in ASCII text format.
- Relays mail between mail servers.

How SMTP Works?

- ① The client connects to the SMTP server.
- ② The client sends HELO/EHLO command.
- ③ The client authenticates (if required).
- ④ The client sends MAIL FROM, RCPT TO and DATA commands.
- ⑤ The server accepts the mail and delivers it to the recipient's server.

What is POP3?

POP3 (Post Office Protocol version 3) is an application layer protocol used by email clients to retrieve emails from a mail server.

Key Features of POP3

- Used for downloading emails from the server to the client.
- Works on TCP port 110 (or 995 for POP3S).
- Downloads emails and (by default) removes them from the server.
- Simple and suitable for offline use.
- Does not synchronize emails across multiple devices.

What is IMAP?

IMAP (Internet Message Access Protocol) is an application layer protocol used by email clients to access and manage emails stored on a mail server.

Key Features of IMAP

- Used for accessing and managing emails on the server.
- Works on TCP port 143 (or 993 for IMAPS).
- Emails remain on the server (not deleted by default).
- Supports folder management (Inbox, Sent, Drafts, etc.).
- Synchronizes emails across multiple devices.

POP3 vs IMAP

Feature	POP3	IMAP
Full Form	Post Office Protocol version 3	Internet Message Access Protocol
Port	110 (995 for POP3S)	143 (993 for IMAPS)
Email Storage	Downloads and removes from server	Keeps emails on server
Access	Best for single device	Best for multiple devices
Folder Management	Not supported	Supported
Synchronization	Not available	Available
Best Used For	Offline use	Online and multi-device use

Summary

SMTP is used for sending emails, POP3 is used for downloading emails to a single device, and IMAP is used for accessing and managing emails on the server across multiple devices.

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Chapter 8 : DNS (Domain Name System)

What is DNS?

DNS (Domain Name System) is an application layer protocol that translates human-readable domain names (like `www.google.com`) into IP addresses (like `142.250.190.78`) that computers use to identify each other on a network.

Key Features of DNS

- Translates domain names to IP addresses.
- Distributed and hierarchical database system.
- Uses UDP port 53 for most queries (TCP also used for zone transfers).
- Caches query results to improve performance.
- Supports various record types (A, AAAA, CNAME, MX, etc.).

How DNS Works?

- ① The user enters a domain name in the browser.
- ② The browser sends a DNS query to the local DNS resolver.
- ③ The resolver checks its cache for the IP address.
- ④ If not found, it queries the Root DNS server.
- ⑤ The Root server responds with the address of the appropriate TLD server.
- ⑥ The resolver queries the TLD server (like `.com`, `.org`).
- ⑦ The TLD server responds with the authoritative DNS server address.
- ⑧ The resolver queries the authoritative DNS server.
- ⑨ The authoritative server responds with the IP address.
- ⑩ The resolver sends the IP address back to the browser.

DNS Query Types

- Recursive Query : The resolver queries on behalf of the client.
- Iterative Query : The server responds with the best information it has.

Common DNS Record Types

Record Type	Description	Example
A	Maps domain name to IPv4 address	<code>example.com</code> → <code>192.168.1.1</code>
AAAA	Maps domain name to IPv6 address	<code>example.com</code> → <code>2001:db8::1</code>
CNAME	Canonical name (alias) of a domain	<code>www</code> → <code>example.com</code>
MX	Mail exchange record	<code>example.com</code> → <code>mail.example.com</code>
NS	Name server record	<code>example.com</code> → <code>ns1.example.com</code>
PTR	Reverse lookup record	<code>1.168.192.in-addr.arpa</code> → <code>example.com</code>
TXT	Text information for the domain	<code>example.com</code> → <code>"v=spf1 include:_spf..."</code>

DNS Message Format

A DNS message consists of the following sections:

- Header Section
- Question Section
- Answer Section
- Authority Section
- Additional Section

Applications of DNS

- Website Browsing
- Email Delivery
- Network Services Discovery
- Load Balancing
- Content Delivery Networks (CDNs)

Summary

DNS is a critical protocol that helps users access websites using easy-to-remember domain names by converting them into IP addresses through a distributed and efficient system.

Networking Protocols

Chapter 9 : DHCP and ARP

What is DHCP?

DHCP (Dynamic Host Configuration Protocol) is a network protocol that automatically assigns IP addresses and other network configuration details to devices on a network.

Key Features of DHCP

- Automatically assigns IP addresses.
- Reduces manual configuration.
- Assigns other network parameters like subnet mask, gateway and DNS.
- Supports IP address lease (temporary assignment).
- Centralized management of IP addresses.

How DHCP Works?

- ① The client broadcasts a DHCP Discover message.
- ② The DHCP server responds with a DHCP Offer.
- ③ The client requests an IP address using DHCP Request.
- ④ The server acknowledges with DHCP Acknowledgement.
- ⑤ The client configures the IP address and other parameters.

What is ARP?

ARP (Address Resolution Protocol) is a network protocol used to map an IP address to a MAC address on a local network.

Key Features of ARP

- Resolves IP address to MAC address.
- Works at the Data Link Layer.
- Used within the local network (LAN).
- Allows communication between devices in the same network.

How ARP Works?

- ① The device checks its ARP cache for the IP to MAC mapping.
- ② If not found, it broadcasts an ARP Request.
- ③ The device with that IP address replies with an ARP Reply (unicast).
- ④ The sender stores the IP to MAC mapping in its ARP cache.
- ⑤ Communication continues using the MAC address.

DHCP Message Types

Message Type	Description
DHCP Discover	Client broadcasts to find available DHCP servers.
DHCP Offer	Server offers an IP address to the client.
DHCP Request	Client requests the offered IP address.
DHCP Acknowledgement	Server confirms the IP address assignment.
DHCP Release	Client releases the IP address when not in use.
DHCP Decline	Client declines the offered IP address.
DHCP Inform	Client requests other configuration details (no IP needed).

Applications of DHCP

- IP address management in large networks
- Reduces IP address conflicts
- Used in offices, schools and organizations
- Supports mobile devices and BYOD environments
- Simplifies network administration

Applications of ARP

- Mapping IP to MAC in local network
- Data transmission within LAN
- Troubleshooting network connectivity
- Used in routers, switches and computers
- Essential for packet delivery in LAN

Summary

DHCP automates IP address assignment and network configuration, while ARP resolves IP addresses to MAC addresses for communication within the local network. Both protocols work together to ensure smooth and efficient network operations.

Networking Protocols

Chapter 10 : TCP and UDP

What is TCP?

TCP (Transmission Control Protocol) is a connection-oriented transport layer protocol that provides reliable, ordered and error-checked delivery of data between applications running on hosts communicating over an IP network.

Key Features of TCP

- Connection-oriented (uses three-way handshake)
- Reliable data transfer with acknowledgements
- Ensures data integrity and in-order delivery
- Flow control and congestion control
- Uses port 20 (FTP data) and 21 (FTP control), 80 (HTTP), 443 (HTTPS), etc.

How TCP Works?

- ① Connection Establishment : Client and server perform a three-way handshake (SYN, SYN-ACK, ACK).
- ② Data Transfer : Data is broken into segments, numbered and sent.
- ③ Acknowledgement : Receiver acknowledges each segment.
- ④ Error Control : Lost segments are retransmitted.
- ⑤ Connection Termination : Connection is closed using FIN and ACK messages.

What is UDP?

UDP (User Datagram Protocol) is a connectionless transport layer protocol that provides fast but unreliable data transfer between applications.

Key Features of UDP

- Connectionless (no handshake)
- No reliability, acknowledgements or retransmissions
- Low overhead and faster than TCP
- Uses port 53 (DNS), 67 (DHCP), 68 (DHCP client), 161 (SNMP), etc.

TCP vs UDP Comparison

Feature	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable (acknowledgements, retransmissions)	Unreliable (no acknowledgements)
Data Delivery	Guaranteed and in-order	Not guaranteed, may be out-of-order
Speed	Slower (higher overhead)	Faster (lower overhead)
Header Size	20 to 60 bytes	8 bytes
Flow Control	Yes	No
Best Used For	File transfer, Web browsing, Email, etc.	Streaming, VoIP, Gaming, DNS, etc.

Common TCP Flags

Flag	Description
SYN	Synchronize sequence numbers (connection request)
SYN-ACK	Synchronize and acknowledge (connection response)
ACK	Acknowledge (confirms data receipt)
FIN	Finish (connection termination)
RST	Reset (abnormal connection termination)

Common Applications

- Applications Using TCP
- Web Browsing (HTTP/HTTPS)
 - Email (SMTP, POP3, IMAP)
 - File Transfer (FTP)
 - Remote Login (SSH, Telnet)
 - Online Banking

- Applications Using UDP
- Video Streaming
 - Online Gaming
 - VoIP (Voice over IP)
 - DNS Queries
 - DHCP (Configuration)

Summary

TCP provides reliable and ordered delivery with higher overhead, suitable for applications where accuracy is important. UDP provides fast and lightweight communication, suitable for time-sensitive applications where speed is more important than reliability.

Networking Protocols

Chapter 11 : IP (Internet Protocol)

What is IP?

IP (Internet Protocol) is a network layer protocol responsible for addressing, routing and delivering packets across multiple networks (internets). It is a connectionless, best-effort protocol.

Key Features of IP

- Logical addressing using IP addresses (IPv4 and IPv6).
- Connectionless and best-effort delivery.
- Routing of packets across multiple networks.
- Fragmentation and reassembly to handle different MTU sizes.
- Protocol independent (works with TCP, UDP, ICMP, etc.).

IP Addressing

- IPv4 : 32-bit address (e.g., 192.168.1.1)
- IPv6 : 128-bit address (e.g., 2001:0db8::1)

IPv4 Address Classes

Class	Leading Bits	First Octet Range	Default Subnet Mask	No. of Networks	No. of Hosts per Network
A	0	1 - 126	255.0.0.0	126	16,777,214
B	10	128 - 191	255.255.0.0	16,384	65,534
C	110	192 - 223	255.255.255.0	2,097,152	254
D (Multicast)	1110	224 - 239	N/A	N/A	N/A
E (Reserved)	1111	240 - 255	N/A	N/A	N/A

IP Datagram Format (IPv4)

0	4	8	16	24	31
Version	IHL	Type of Service	Total Length		
Identification			Flags	Fragment Offset	
Time to Live (TTL)		Protocol	Header Checksum		
Source IP Address					
Destination IP Address					
Options (if any)				Padding	
Data (Payload)					

Field Description

- Version : IP version (4 for IPv4).
- IHL : Internet Header Length.
- Total Length : Header + data length.
- Identification : Used for fragmentation.
- Flags : Control fragmentation.
- Fragment Offset : Position of fragment.
- TTL : Prevents infinite packet looping.
- Protocol : Next layer protocol (TCP=6, UDP=17).
- Header Checksum : Error checking.
- Source/Destination IP : Sender/Receiver.

How IP Works?

- ① The source creates an IP datagram with source and destination IP addresses.
- ② The datagram is routed through intermediate routers based on routing tables.
- ③ Each router decreases the TTL and forwards the datagram toward the destination.
- ④ If TTL becomes 0, the datagram is discarded and an ICMP Time Exceeded message is sent.
- ⑤ The destination receives the datagram and passes the payload to the appropriate protocol.

IP Fragmentation

- If a datagram is larger than the MTU of a network, it is fragmented.
- Each fragment is routed independently and reassembled at the destination.
- Fragment identification, flags and offset help in reassembly.

IP vs TCP (Quick Comparison)

Feature	IP	TCP
Layer	Network Layer	Transport Layer
Connection	Connectionless	Connection-oriented
Reliability	Best-effort	Reliable
Data Unit	Datagram	Segment
Error Control	No	Yes
Flow Control	No	Yes
Speed	Faster	Slower (overhead)
Best Used For	Routing across networks	Reliable data transfer

Applications of IP

- Internet communication and routing
- Web browsing
- Video streaming
- Email delivery
- Online gaming
- Voice over IP (VoIP)
- IoT and remote devices communication

Summary

IP is the backbone of the internet, enabling packets to travel across multiple networks using logical addressing and routing. It provides best-effort delivery, while protocols like TCP ensure reliable communication when required.

Networking Protocols

Chapter 11 : IP (Internet Protocol)

What is IP?

IP (Internet Protocol) is a network layer protocol responsible for addressing and routing packets across multiple networks.

Key Features of IP

- Provides logical addressing using IP addresses
- Connectionless and best-effort delivery
- Supports fragmentation and reassembly
- Works across heterogeneous networks
- IPv4 (32-bit) and IPv6 (128-bit) addressing

IP Header Format (IPv4)

Version (4)	IHL (4)	Type of Service (8)	Total Length (16)
Identification (16)		Flags (3)	Fragment Offset (13)
Time to Live (8)		Protocol (8)	Header Checksum (16)
Source IP Address (32 bits)			
Destination IP Address (32 bits)			
Options (if any) + Padding			
Data (Payload)			

IPv4 Addressing

- IPv4 address is 32 bits (4 octets).
- Written in dotted-decimal form (e.g., 192.168.1.1).
- Address is divided into Network ID and Host ID.
- Classes: A, B, C, D, E (legacy).
- Private IP ranges:
 - 10.0.0.0 - 10.255.255.255 (Class A)
 - 172.16.0.0 - 172.31.255.255 (Class B)
 - 192.168.0.0 - 192.168.255.255 (Class C)

IP Address Classes (Legacy)

Class	First Bits	Default Subnet Mask	# Networks	# Hosts / Network
A	0	255.0.0.0	126	16,777,214
B	10	255.255.0.0	16,384	65,534
C	110	255.255.255.0	2,097,152	254
D	1110	Multicast (No mask)	-	-
E	11110	Reserved	-	-

IP Address Types

- Unicast : One-to-one communication
- Broadcast : One-to-all in the same network
- Multicast : One-to-many (group of hosts)
- Anycast : One-to-nearest (used in routing)

IPv4 Packet Flow



IPv6 (Internet Protocol version 6)

- IPv6 uses 128-bit addresses (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334)
- Provides a much larger address space
- Supports auto-configuration and better security
- No broadcast - uses multicast instead

IPv6 Packet Structure

Version (4)	Traffic Class (8)	Flow Label (20)
Payload Length (16)		Next Header (8)
		Hop Limit (8)
Source Address (128 bits)		
Destination Address (128 bits)		
Payload (Data)		

Common IP Utilities (Commands)

Command	Purpose
ping	Test reachability to a host
tracert / traceroute	Show path to destination
ipconfig / ifconfig	Display IP configuration
nslookup	Query DNS information
netstat	Show network connections

Fragmentation and Reassembly

- IP packets may be fragmented if larger than MTU.
- Each fragment has Identification, Flags and Offset.
- Reassembly happens at the destination host.
- Too many fragments can affect performance.

Summary

IP is the foundation of internetworking. It provides logical addressing, routing and delivers packets across networks in a best-effort manner. IPv4 is widely used, while IPv6 is the next generation to support the growing number of devices and services.

Networking Protocols

Chapter 12 : Application Layer Protocols

What is Application Layer?

The Application Layer is the topmost layer in the TCP/IP model. It provides network services directly to user applications. Application layer protocols define how applications format messages, exchange information and perform specific network operations.

Key Features of Application Layer Protocols

- Provide services to end-user applications.
- Define message formats and communication rules.
- Support a wide range of user services (web, email, file transfer, name resolution, etc.).
- May use either TCP or UDP as the transport protocol.
- Examples: HTTP, FTP, SMTP, DNS, POP3, IMAP, SNMP, DHCP, Telnet, SSH, etc.

Common Application Layer Protocols

Protocol	Use	Transport Protocol	Port Number	Description
HTTP	Web browsing	TCP	80	Transfers hypertext data on the web.
HTTPS	Secure web browsing	TCP	443	HTTP over SSL/TLS for secure communication.
FTP	File transfer	TCP	20, 21	Transfers files between client and server.
SMTP	Email sending	TCP	25	Sends email from client to mail server.
POP3	Email retrieval	TCP	110	Downloads email from server to client.
IMAP	Email access	TCP	143	Accesses email on server (sync across devices).
DNS	Name resolution	UDP/TCP	53	Resolves domain names to IP addresses.
SNMP	Network management	UDP	161	Monitors and manages network devices.
DHCP	IP address assignment	UDP	67 (server) 68 (client)	Dynamically assigns IP addresses to clients.
Telnet	Remote login (insecure)	TCP	23	Provides remote terminal access (unencrypted).
SSH	Secure remote login	TCP	22	Secure remote terminal access.

How Application Layer Protocols Work?

- ① The user application sends a request to the application layer.
- ② The application layer formats the request according to the protocol rules.
- ③ The request is passed to the transport layer (TCP/UDP).
- ④ The transport layer delivers the request to the destination application.
- ⑤ The destination application processes the message and sends a response.

Example: Web Browsing (HTTP)

- ① User enters a URL in the browser.
- ② Browser extracts the domain name and uses DNS to get the IP address.
- ③ Browser sends an HTTP request to the web server using TCP (port 80).
- ④ Web server processes the request and sends an HTTP response.
- ⑤ Browser receives the response and displays the web page.
- ⑥ Connection is closed.

Applications of Application Layer Protocols

- Web browsing (HTTP/HTTPS)
- File transfer (FTP)
- Email services (SMTP, POP3, IMAP)
- Domain name resolution (DNS)
- Network management (SNMP)
- Remote access (Telnet, SSH)
- Dynamic IP assignment (DHCP)

Security Considerations

- Use secure protocols (HTTPS, SSH) whenever possible.
- Avoid sending sensitive data over unencrypted protocols (HTTP, Telnet).
- Keep software and protocols updated.
- Use strong authentication for remote access.
- Monitor and limit open ports.

Summary

Application layer protocols enable user applications to communicate over the network by defining the rules for data exchange and services. These protocols are essential for providing services like web browsing, email, file transfer, and network management.